Total No. of Questions : 8]

SEAT No. :

PE-4292

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S.E. (AI & ML)

FUNDAMENTALS OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

(2019 Pattern) (Semester - IV) (218553)

Time : 2½ Hours] *[Max. Marks : 70*

Instructions to the candidates:

1) ***Answer Q.1 or Q.2, Q.3 or Q.4, Q.5 or Q.6, Q.7 or Q.8.***
2) ***Neat diagrams must be drawn wherever necessary.***
3) ***Figures to the right side indicate full marks.***
4) ***Assume suitable data if necessary.***

Q1) a) Discuss various approaches and issues in knowledge representation. Also discuss various problems occur in representing knowledge? **[6]**

b) Expalin the Unification used for reasoning under predicate logic with example. **[6]**

c) Compare between Forward Chaining and Backward Chaining. **[6]**

OR

Q2) a) Explain the algorithm of predicate logic resolution. **[6]**

b) What is reasoning in AI? Explain its types? **[6]**

c) Difference between declarative and procedural knowledge. **[6]**

Q3) a) Explain supervised and unsupervised and reinforcement learning with example. **[8]**

b) Explain the data preprocessing steps in machine learning. **[6]**

c) Distinguish between artificial intelligence and machine learning. **[3]**

OR

Q4) a) Differentiate between positive class and negative class. Explain the training versus testing phase in machine learning. **[4]**

b) Explain different cross validation techniques. **[8]**

c) Discuss different applications of machine learning. **[5]**

P.T.O.

Q5) a) Highlighted differences between supervised, unsupervised and semi-supervised learning in machine learning? Explain with examples. **[9]**

b) Define Principal Component Analysis (PCA) and explain how it reduces the dimensionality of data. **[9]**

OR

Q6) a) Define supervised learning and provide real life examples where it is commonly applied. **[9]**

b) Write a short note on Dimensionality reduction. **[9]**

Q7) a) How the performance of Regression is Assessed? Write any four-performance metrics used for Regression. **[7]**

b) Consider the three-class classification matrix. Calculate accuracy, precision and recall per class. **[10]**

Confusion Matrix		Predicted		
Actual	A	8	10	1
	B	5	60	50
	C	3	30	200

OR

Q8) a) Define following terms : **[8]**

i) Accuracy

ii) Precision

iii) Recall

iv) Error rate

b) What is difference between simple linear and multiple linear regressions? Explain how to assessign performance of Regression. **[9]**

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